Did the 2022–2024 rate shock render IRRBB compliance and NII preservation incompatible for retail banks?

Nino Petiard

Master MBFA - Paris 1 Panthéon Sorbonne

2025-2026

Contents

Glossary **1**

1 Introduction **2**

2 Data, calibration and methodology **4**
2.1 The legacy of the Great Moderation: genesis of a balance sheet structurally exposed to interest rate risk 4
2.2 The bank balance sheet: calibration and structural diagnostic . . . 5
2.3 The yield curve and its inversion . . . 6
2.4 Client behaviour modelling . . . 7

3 Interest rate risk exposure diagnostic **8**
3.1 The rate gap: a radiograph of transformation . . . 8
3.2 Net interest income: a misleading short-term resilience . . . 9
3.3 Economic value of equity: the true alarm signal . . . 10

4 Hedging strategy and NII/EVE efficient frontier **11**
4.1 Calibrating the hedging portfolio: a bucket-by-bucket approach . . . 11
4.2 The NII/EVE efficient frontier . . . 12
4.3 Carry cost and the macroeconomic window: the cornerstone of compatibility . . . 14
4.4 Sensitivity to behavioural assumptions . . . 14

5 Extensions: towards a stochastic efficient frontier **16**
5.1 From binary to probabilistic compliance: Monte Carlo simulation using the Hull-White model . . . 16
5.1.1 Principle and model justification . . . 16
5.1.2 Simulation protocol . . . 17
5.1.3 Results . . . 17
5.2 Endogenous client behaviours and compliance robustness: a dynamic Monte Carlo loop . . . 18
5.2.1 Motivation . . . 18
5.2.2 Dynamic behavioural functions . . . 18
5.2.3 Results . . . 19

6 Conclusion **21**
6.1 Answer to the research question . . . 21
6.2 Perspectives and limitations . . . 21

Glossary **23**

References **26**

Introduction

Between July 2022 and September 2023, the European Central Bank raised its deposit rate from −0.50% to 4.00%, a 450 basis point increase over ten consecutive hikes, the fastest and largest tightening since the creation of the euro area.[1] This shock brought to light a structural vulnerability that had long lain dormant in the European banking sector: *maturity transformation*.

Maturity transformation is the founding activity of retail banking. It consists of collecting short-term resources (demand deposits, savings accounts) and deploying them in long-term assets (mortgage loans, sovereign bonds). This *duration* asymmetry is the primary source of bank net interest income, but it mechanically exposes the institution to interest rate risk in the *banking book*, referred to by the acronym **IRRBB** (*Interest Rate Risk in the Banking Book*). This risk manifests itself in two complementary and often antagonistic forms, which constitute the two analytical pillars of this project.

Net Interest Income (**NII**) measures the impact of a rate shock on net interest flows in the short term, typically over a one-to-three-year horizon. It is the CFO's metric: it captures the immediate effect on the income statement. **Economic Value of Equity** (**EVE**), on the other hand, measures the impact of the same shock on the net present value of all future balance sheet cash flows over their remaining life. It is the risk manager's metric: it captures the latent destruction of economic value, regardless of its accounting realisation. Both measures are required by the IRRBB regulatory framework of the EBA[2] and the Basel Committee,[3] with respective prudential alert thresholds of 15% of *Tier 1* capital[4] for $|\Delta\text{EVE}|$ and 2.5% of total assets for $|\Delta\text{NII}|$.

The tension between NII and EVE is the central dilemma of *asset-liability management*. Protecting EVE against a rate rise requires hedging instruments, typically *pay-fixed interest rate swaps*,[5] which generate a cost on NII in a stable or falling rate scenario. Conversely, maximising NII by forgoing hedging exposes the institution to massive economic value losses and prudential alerts.

French retail banks illustrate this tension with particular acuity. Between 2015 and 2022, they had massively extended fixed-rate mortgage loans at historically low levels,[6] benefiting from a prolonged accommodative monetary policy. These assets, whose duration can reach fifteen to eighteen years, constitute the bulk of interest rate risk in the banking book. Their economic value collapses when rates rise sharply, even if contractual cash flows are assured and nothing in the income statement immediately reflects it. It is precisely this dissociation between latent and realised losses that lies at the heart of this work: an EVE loss may remain invisible in accounting terms as long as liquidity is preserved, but it exists economically from the first day of the shock. The French specificity, namely the near-universal use of fixed-rate mortgage lending, places French banks in a structurally more exposed position than their German or Scandinavian counterparts, whose lending models assign a much greater share to variable-rate products. Several institutions illustrated this in their *Pillar 3* documents,[7] disclosing significant EVE variations under regulatory stress scenarios.[8]

[1] ECB, *Key ECB interest rates*, historical data, 2024. https://www.ecb.europa.eu/stats/policy_and_exchange_rates/key_ecb_interest_rates/html/index.en.html.

[2] The *European Banking Authority* (EBA) is the European banking supervisory authority responsible for harmonising prudential practices within the European Union. EBA, *Guidelines on the management of interest rate risk arising from non-trading book activities*, EBA/GL/2022/14, October 2022.

[3] BCBS, *Interest rate risk in the banking book*, standards, April 2016. https://www.bis.org/bcbs/publ/d368.htm.

[4] A $|\Delta\text{EVE}|$ exceeding 15% of Tier 1 signals that the rate shock would destroy more than one seventh of the institution's loss-absorption capacity. The $|\Delta\text{NII}|$ threshold of 2.5% of total assets reflects a significant compression of current income relative to the size of the activity. These thresholds are not strict regulatory limits but supervisory indicators, beyond which the supervisor may require corrective measures.

[5] In the event of a rate rise, the received floating leg increases while the paid fixed leg remains constant, generating a gain that offsets the loss in economic value of the balance sheet.

[6] Banque de France, *Stat Info, Production de crédits immobiliers*, 2022. Average rates on new mortgage loans reached a low of 1.06% in December 2021. https://www.banque-france.fr/statistiques/taux-et-cours/taux-des-credits.

[7] The *Pillar 3* refers to the third pillar of the Basel III prudential framework, dedicated to market discipline through transparency. It requires credit institutions to publish quantitative and qualitative information on their risk profile, including their IRRBB exposure and EVE variations under regulatory stress scenarios.

[8] By way of example, Crédit Agricole published in its 2023 annual report an EVE sensitivity in the order of several billion euros under the parallel upward shock of 200 bp. https://www.credit-agricole.com/finance/publications-financieres.

This is where the tension crystallises. Bringing EVE back within regulatory limits requires hedging, and hedging comes at a cost to NII. This dilemma is not new in theory, but the 2022–2024 shock made it operationally acute for institutions that had spent ten years in a near-zero rate environment where, as we show in Section 2, under-hedging was rational. To this is added an unusual feature: the inverted yield curve of March 2024 structurally changes the *carry* of hedging instruments, in a way that is favourable but transitory.

The question that follows is:

Did the 2022–2024 rate shock render IRRBB compliance and NII preservation incompatible for retail banks?

To answer this question, this work constructs a complete quantitative simulation based on a stylised French retail bank balance sheet of ten billion euros, calibrated as of March 2024. The approach follows the operational logic of an ALM team: calibration of the reference yield curve, balance sheet construction and client behaviour modelling, calculation of NII and EVE under EBA scenarios, analysis of the *rate gap*, implementation of a bucket-by-bucket swap hedging strategy, and construction of the *NII/EVE efficient frontier*. Each step is accompanied by a critical discussion of assumptions and their economic implications. Two complementary analyses enrich the central diagnostic: a formal sensitivity analysis with respect to behavioural assumptions, and a quantification of the carry cost of swaps in the context of the inverted curve.

The paper is structured as follows. Section 2 lays the foundations: the genesis of the exposed balance sheet, calibration, and the yield curve. Section 3 establishes the exposure diagnostic via the rate gap, NII, and EVE under the six regulatory scenarios. Section 4 constructs the hedging strategy and traces the NII/EVE efficient frontier, quantifying the decisive role of carry. Section 5 enriches this frontier through two extensions: a Monte Carlo simulation using a Hull-White model, which probabilises compliance, and an integration of dynamic client behaviours, which tests the robustness of conclusions when the balance sheet adapts to the rate path. Section 6 synthesises the results and concludes.

Key results. Without hedging, four out of six EBA scenarios trigger a prudential alert on EVE, with $|\Delta\text{EVE}|$ reaching 37.3% of Tier 1 under a parallel upward shock of 200 bp, i.e., 2.5 times the regulatory threshold. An IRS swap hedge representing 33% of the balance sheet, calibrated through formal optimisation under regulatory constraints, brings this figure within regulatory limits for all six EBA scenarios, while generating a positive carry of 29.8 M€ per year thanks to the inverted curve configuration of March 2024, making hedging simultaneously a regulatory necessity and economically attractive. The NII/EVE efficient frontier reveals a compatibility zone between 50% and 110% of hedge coverage. The Hull-White Monte Carlo simulation calibrated on the March 2024 OIS curve specifies that this zone is accessible with a probability of 83.3% under the distribution of plausible trajectories. This window is however transitory: the normalisation of rates will mechanically push carry into negative territory from 2025 onwards.

Data, calibration and methodology

This section lays the empirical and methodological foundations of the simulation. It follows a four-step progression: understanding why French bank balance sheets displayed such a marked risk profile in March 2024, constructing and calibrating the stylised balance sheet that will serve as the basis for all calculations, establishing the reference yield curve, and modelling client behaviours upon which the precision of ALM metrics depends. Each step conditions the next: without this genealogy, the balance sheet would be an arbitrary object; without a calibrated balance sheet, rate scenarios would have no economic substrate; without behavioural assumptions, NII and EVE calculations would be mechanically incorrect.

The legacy of the Great Moderation: genesis of a balance sheet structurally exposed to interest rate risk

The risk profile of French retail banks in March 2024 is not accidental: it is the direct product of a decade of incentives consistent with a near-zero rate environment. Understanding this genealogy is necessary before any calibration, as it conditions all structural choices in the modelled balance sheet and explains why the observed under-hedging was not a management error, but a rational response to a given environment.

This low-rate environment was not a cyclical anomaly but the product of deep structural forces widely anticipated to be lasting. Borio and White (2004) documented the emergence of a global macroeconomic regime characterised by durably contained inflation, enhanced central bank credibility, and a moderation of economic cycle volatility.[9] This regime, sometimes referred to as the *Great Moderation*, had anchored long-rate expectations at structurally low levels: markets anticipated no significant inflationary resurgence, and central banks appeared to have permanently tamed the price cycle. In this context, holding long-term fixed-rate assets was not only profitable but appeared to carry little risk: if rates were not expected to rise, the long duration of assets was not a vulnerability but a stable source of margin.

It is within this framework that the commercial logic of French retail banks developed. Three converging factors shaped their balance sheet profile. First, the cost of hedging: covering via a ten-year pay-fixed IRS meant paying a market fixed rate while receiving a near-zero Euribor, generating a net charge of 20 to 45 M€ per year on a portfolio of 2,000 to 3,000 M€, equivalent to 30 to 40% of the total gross margin on a mortgage portfolio whose rates were already scraping the floor at 1%.[10] In such an environment, extending fixed-rate mortgages over fifteen to twenty years and funding them with demand deposits or short-term resources was the dominant strategy: the yield curve was normally upward-sloping, long rates exceeded short rates, and the *transformation margin* constituted a near-guaranteed rent. The incentive to lengthen asset duration while maintaining short, flexible liabilities was therefore structural, and hedging this transformation risk meant forgoing this rent for no apparent benefit. Hedging thus amounted to significantly impairing operational profitability to protect against a risk that neither markets nor supervisors anticipated in the near term. Second, the supervisory framework exerted no effective pressure in this direction: the EBA's IRRBB guidelines did not acquire their binding format until 2022, and a 200 bp shock applied to near-zero rates simply did not trigger prudential alerts.[11] Finally, bank executives, assessed on annual profitability metrics (ROE, cost-to-income ratio), had no incentive to immediately impair these indicators to protect against a hypothetical shock, a form of *disaster myopia*[12] well documented in the literature.

The result is a typical balance sheet profile: a high proportion of long-duration fixed-rate assets, rapidly repricing liabilities, insufficient hedging. This is not an anomaly but the archetype of the French retail bank built during the decade of low rates, whose asset allocation choices were perfectly consistent with the dominant expectations of the

[9] Borio, C., White, W., *Whither monetary and financial stability? The implications of evolving policy regimes*, BIS Working Paper No. 147, 2004. https://www.bis.org/publ/work147.pdf.

[10] Observatoire Crédit Logement / CSA, *Monthly dashboard of the mortgage market*, 2019–2021 data. www.lobservatoirecreditlogement.fr.

[11] ECB, *Supervisory Review and Evaluation Process (SREP), Methodology Booklet*, 2019.

[12] Guttentag, J., Herring, R., *Disaster Myopia in International Banking*, Essays in International Finance, Princeton University, 1986. The authors show that agents systematically underweight low-probability risks when these have not materialised recently.

time. It is precisely this profile that the stylised balance sheet in the next section seeks to reproduce.

The bank balance sheet: calibration and structural diagnostic

The modelled balance sheet represents a mid-sized French retail bank with a total balance sheet of ten billion euros. It is a stylised balance sheet, constructed for analytical purposes: its parameters are calibrated on publicly available data for this category of institutions, notably Banque de France banking statistics,[13] data from the Observatoire Crédit Logement[14] and ACPR statistical reports.[15] It does not purport to faithfully reproduce the balance sheet of any particular institution. The objective is to have an economically coherent substrate, sufficiently representative of the sector for conclusions to be generalisable, while remaining simple enough for the mechanisms to be legible.

Table 1: Balance sheet structure — assets (total: 10,000 M€)

Item	Amount	Rate	Maturity	Type
Fixed-rate mortgage loans	3,500 M€	1.80%	18 yrs	Fixed
Variable-rate mortgage loans	1,000 M€	4.10%	15 yrs	Variable
Consumer loans	800 M€	5.50%	3 yrs	Fixed
Fixed-rate corporate loans	1,500 M€	2.50%	5 yrs	Fixed
Variable-rate corporate loans	700 M€	4.35%	3 yrs	Variable
Sovereign bond portfolio	1,500 M€	1.20%	6 yrs	Fixed
Treasury and liquid assets	1,000 M€	3.90%	0.25 yr	Variable

Table 2: Balance sheet structure — liabilities (total: 10,000 M€)

Item	Amount	Rate	Conv. maturity	Type
Demand deposits	3,500 M€	0.10%	3 yrs	Variable
Livret A and regulated savings	2,000 M€	3.00%	4 yrs	Administered
Term deposits	1,500 M€	3.50%	1.5 yrs	Fixed
Senior bonds issued	1,500 M€	1.80%	4.5 yrs	Fixed
ECB refinancing (TLTRO)	600 M€	0.75%	0.5 yr	Fixed
Tier 1 equity	900 M€	—	—	—

Four structural indicators summarise the risk profile of this balance sheet and verify that it faithfully reproduces the typical behaviour described in the previous section. The net gross margin stands at 1.25% (average asset rate of 2.73% versus average liability rate of 1.48%), which is representative of French retail banks over the period considered.[16] The maturity gap is 7.1 years (9.9 years on the asset side versus 2.8 years on the liability side), reflecting significant maturity transformation. The share of fixed-rate assets reaches 73%, a threshold beyond which EVE sensitivity to *parallel shocks* becomes particularly strong, as shown by Entrop et al. (2012).[17]

The EVE in the central scenario stands at 1,024.9 M€, slightly above the book equity of 900 M€, reflecting the franchise value of the commercial portfolio.

These four indicators converge to characterise the balance sheet as structurally *liability-sensitive*: liabilities *reprice* much faster than assets. Demand deposits and short-term resources follow monetary policy decisions almost immediately, while fixed-rate mortgages extended for fifteen to twenty years remain locked at their original rate for their entire life. When rates rise, liability interest charges therefore increase before asset interest revenues adjust, mechanically compressing the net interest margin, exactly the scenario that played out between 2022 and 2024.

[13] Banque de France, *Les établissements de crédit en France, statistiques*, 2023 data. https://www.banque-france.fr/fr/publications-et-statistiques/statistiques.

[14] Observatoire Crédit Logement / CSA, *Monthly dashboard of the mortgage market*, January 2024. www.lobservatoirecreditlogement.fr.

[15] ACPR, *Les chiffres du marché français de la banque et de l'assurance*, annual statistical report, 2023 data. https://acpr.banque-france.fr/system/files/2025-04/20241024_acpr_rapport_chiffres_2023.pdf.

[16] Banque de France, *Les établissements de crédit en France, statistiques*, 2023 data. Net margins of French retail banks oscillated between 0.9% and 1.3% in the first quarter of 2024. https://www.banque-france.fr/fr/publications-et-statistiques/statistiques.

[17] Entrop, O., Memmel, C., Ruprecht, B., Wilkens, M., *Determinants of bank interest margins: Impact of maturity transformation*, *Journal of Banking & Finance*, 54, 1–19. https://doi.org/10.1016/j.jbankfin.2014.12.001.

The representativeness of this balance sheet is confirmed by the public data of major French institutions. In their Pillar 3 reports published in 2023, BNP Paribas disclosed a $|\Delta EVE|$ of approximately 2.2 billion euros under the parallel upward shock of 200 bp, representing approximately 9% of its consolidated Tier 1, and Société Générale reported an EVE sensitivity of 1.4 billion euros under the same scenario, approximately 7% of its Tier 1.[18] These magnitudes, expressed relative to the total balance sheet of these institutions, are consistent with the $|\Delta EVE|$ of 37.3% of Tier 1 obtained on our stylised balance sheet, which deliberately reflects the profile of a mid-sized institution less diversified and therefore more concentrated in fixed-rate mortgages than large universal banks. It is on this balance sheet that the entire simulation rests; it remains to define the rate environment in which it will be subjected to stress scenarios.

The yield curve and its inversion

The simulation uses the EUR OIS (*Overnight Index Swap*) swap curve of March 2024, preferred over the OAT curve, which incorporates a sovereign spread irrelevant to the assessment of the intrinsic interest rate risk of the balance sheet.[19]

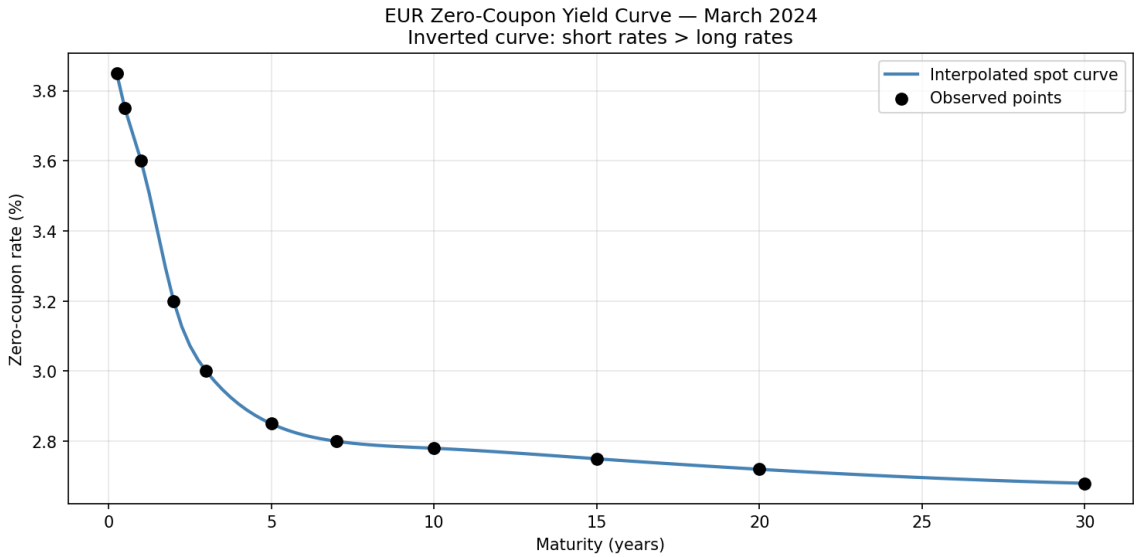

Figure 1: EUR zero-coupon yield curve, March 2024 (source: EUR mid-market swap rates, Bloomberg). The inverted configuration reflects market anticipation of an upcoming rate cut cycle.[20]

The defining feature of this curve is its inversion: the three-month rate stands at 3.85% while the ten-year rate is only 2.78%, an inversion of 107 basis points between these two maturities, and of 117 basis points relative to the thirty-year rate (2.68%). This configuration is historically rare. According to Campbell and Shiller (1991),[21] an inverted curve predicts with high historical reliability an economic slowdown and a future rate cut cycle: the market anticipated in March 2024 that the elevated level of short rates was transitory. For a retail bank structurally borrowing short and lending long, this configuration is doubly unfavourable: it immediately compresses the transformation margin since the cost of short-term resources exceeds the return on long-term assets, and it creates an adverse reinvestment risk when fixed-rate loans mature in a potentially lower rate environment.

This inverted curve context nonetheless presents a counter-intuitive implication for the hedging strategy, analysed in Section 4: long swap rates (2.81% on ten years) are lower than short rates (Euribor 6M at 3.75%), generating

[18] BNP Paribas, *Rapport sur les risques — Pilier 3*, FY2023, IRRBB sensitivity table, p. 214. https://invest.bnpparibas/document/document-denregistrement-universel-et-rapport-financier-annuel-2023-pdf Société Générale, *Document d'enregistrement universel 2024*, section 4.6 Global interest rate risk, p. 312. https://www.societegenerale.com/sites/default/files/documents/2024-03/document-enregistrement-universel-2024.pdf.

[19] The OIS curve is indexed on the €STR (Euro Short-Term Rate), the ECB overnight rate free of interbank credit risk. It has constituted the risk-free reference curve since the reform of rate benchmarks in 2021 (BMR regulation, EU 2016/1011). https://www.ecb.europa.eu/stats/financial_markets_and_interest_rates/euro_short-term_rate/html/index.en.html.

[20] Interpolation between the eleven observed points is performed using a cubic spline (the `CubicSpline` class from `scipy.interpolate`), ensuring continuity of the first derivative, a necessary condition for the correct calculation of duration sensitivities.

[21] Campbell, J.Y., Shiller, R.J., *Yield Spreads and Interest Rate Movements: A Bird's Eye View*, *Review of Economic Studies*, 58(3), 1991, pp. 495–514. https://doi.org/10.2307/2298008.

an unusually favourable positive carry for a pay-fixed IRS. This configuration structurally changes the cost-benefit calculation of hedging: the same instruments that cost 30 to 40 M€/year in negative carry before 2022 now generate a net revenue. It is in this rate context, simultaneously constraining for the balance sheet and exceptionally favourable for hedging, that the modelling of client behaviour below takes place.

Client behaviour modelling

The precision of ALM calculations depends critically on the modelling of two client behaviours that deviate from apparent contractual terms: the true economic duration of demand deposits, and the prepayment rate on fixed-rate mortgages. These assumptions directly condition the magnitude of the rate gap, and therefore of the NII and EVE calculated in the following section.

The approach follows a two-step progression: we first set the static assumptions required by the EBA regulatory framework,[22] which serve as the basis for the central diagnostic in Sections 3 and 4, and then enrich them in Section 5 with dynamic behavioural functions integrated into a Monte Carlo simulation.

Demand deposits. Although legally repayable at any time, demand deposits have a significantly longer economic duration, due to switching costs and loyalty effects documented by Sharpe (1990).[23] EBA guidelines impose a conventional duration capped at five years.[24] The duration retained for the baseline diagnostic is three years, the most common choice in French market practice, with a 70%/30% segmentation between stable and volatile deposits.[25] This fixed-duration assumption has however a limitation: it freezes depositor behaviour independently of the rate level, whereas Memmel (2011) shows that more mature institutions often retain shorter durations, calibrated on historical episodes of deposit volatility.[26] It is precisely this limitation that the dynamic extension of Section 5 addresses, by making the duration of *demand deposits* endogenous to the absolute level of short rates.

Prepayments. The *prepayment* rate depends on the spread between market and contractual rates, a relationship formalised by Richard and Roll (1989) on US MBS.[27] In March 2024, the market rate stands at 4.20%, i.e., 240 bp above the average contractual rate of 1.80%[28]: no rational borrower will refinance their loan to take out a new one at more than twice the current rate, so the *lock-in* effect is maximal and observed prepayments are limited to constrained repayments (relocation, divorce, death) independent of rate conditions. The retained floor rate is therefore 5% per year.[29]

[22] The static approach, known as *run-off*, is one of the two approaches mandated by the EBA/GL/2022/14 guidelines for the calculation of NII under stress. It consists of not renewing outstanding amounts maturing, thereby ensuring comparability between institutions and forming the common methodological basis for all IRRBB regulatory calculations.

[23] Sharpe, S.A., *Asymmetric information, bank lending and implicit contracts: A stylized model of customer relationships*, *Journal of Finance*, 45(4), 1990, pp. 1069–1087. https://doi.org/10.1111/j.1540-6261.1990.tb02427.x.

[24] EBA/GL/2022/14, October 2022, paragraph 115.

[25] Consistent with the *core/non-core* distinction of paragraphs 116–121 of EBA/GL/2022/14.

[26] Memmel, C., *Banks' exposure to interest rate risk*, Deutsche Bundesbank, Discussion Paper 07/2011. https://doi.org/10.1016/j.jbankfin.2010.08.003.

[27] Richard, S.F., Roll, R., *Prepayments on fixed-rate mortgage-backed securities*, *Journal of Portfolio Management*, 15(3), 1989, pp. 73–82.

[28] Observatoire Crédit Logement / CSA, *Monthly dashboard*, January 2024. www.lobservatoirecreditlogement.fr.

[29] Consistent with the 5–7% observed on the French stock in 2023, versus 15–20% during the refinancing waves of 2016–2017. Banque de France, *Stat Info*, 2023. https://www.banque-france.fr/statistiques/taux-et-cours/taux-des-credits.

Interest rate risk exposure diagnostic

This diagnostic follows the operational logic of an ALM team: starting from the balance sheet to locate risk via the rate gap, translating it into an impact on the income statement through NII, then measuring the full economic cost via EVE. The three metrics are calculated by applying the six EBA regulatory stress scenarios to the EUR OIS swap curve of March 2024, taken as the base rate.[30] These scenarios do not reproduce the historical 2022–2024 shock; they test the resilience of this balance sheet, as it stands in March 2024, to further deformations of the curve: this is the operational question that an ALM manager asks in committee.

The rate gap: a radiograph of transformation

The rate gap measures, for each *maturity bucket* defined by EBA standards, the difference between outstanding assets and liabilities liable to reprice within that period. It is the fundamental diagnostic tool of ALM, which quantifies the temporal asymmetry inherent in maturity transformation.

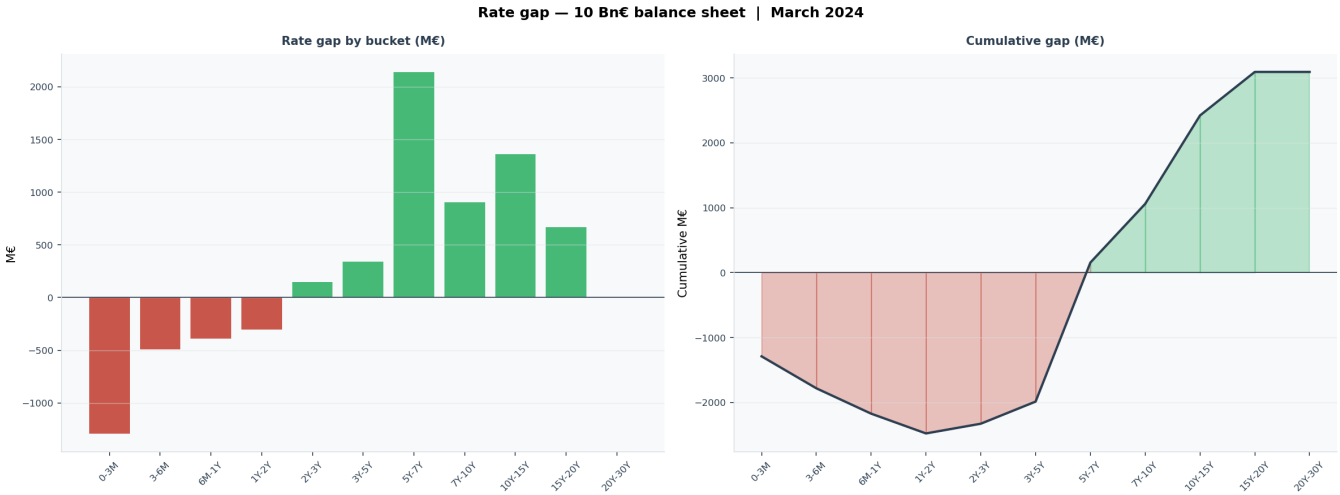

Figure 2: Rate gap by bucket (left) and cumulative gap (right). Red bars signal buckets in *liability-sensitive* positions; the coloured area of the cumulative chart materialises the amplitude of net exposure.

Results reveal a massively *liability-sensitive* position on short buckets. The 0–3 month bucket records a negative gap of −1,290 M€: liabilities repricing this quarter represent 4,258 M€, against only 2,969 M€ of assets. This imbalance persists up to the 1–2 year bucket, bringing the cumulative gap to −2,477 M€. The situation reverses from the 2–3 year bucket, as the first maturities of fixed-rate loans feed the asset side without corresponding liabilities. The position becomes significantly *asset-sensitive* on the 5–7 year buckets (+2,142 M€) and 10–15 years (+1,361 M€), precisely where long-term mortgage loans and the sovereign bond portfolio are concentrated. The final cumulative gap converges towards +3,091 M€ on the last bucket.

The economic translation is direct: a 200 bp upward shock on short rates generates additional interest charges on net repriceable liabilities over a two-year horizon of approximately $2,477 \times 2\% \approx 49$ M€ per year, consistent with the ΔNII calculated in the next section. The concentration of the asset-sensitive gap on the 5–20 year buckets constitutes the substrate of EVE exposure: it is precisely on these long maturities that curve deformation generates massive present value losses, as revealed in Section 3.3.

[30] EBA/GL/2022/14, paragraphs 105–112. The six scenarios are: *parallel up* and *parallel down* (uniform shift of the entire curve by ±200 bp); *steepener* (rising long rates, falling short rates); *flattener* (falling long rates, rising short rates); *short rate up* and *short rate down* (±250 bp shock on short rates only, without shifting long rates).

Net interest income: a misleading short-term resilience

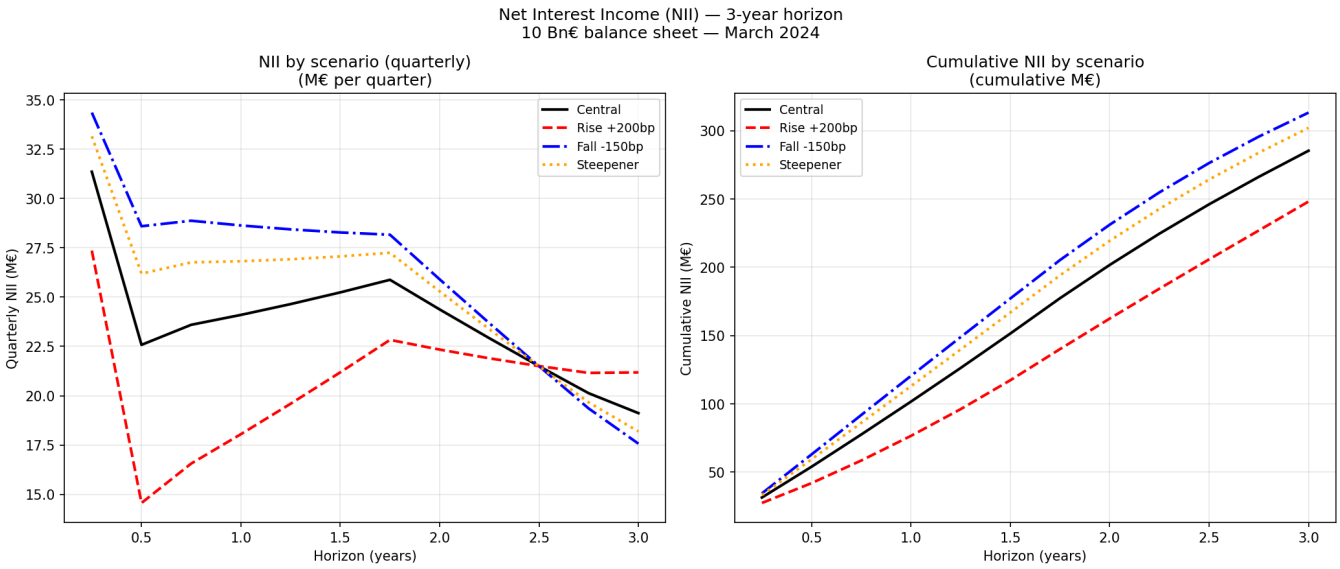

Figure 3: Semi-annual and cumulative NII over three years under four rate scenarios. The central scenario generates 285.4 M€ of NII over the period. The 200 bp rise compresses NII by 37.2 M€ (−13.0%).

The central scenario corresponds to a freeze of the yield curve at its March 2024 level over the entire simulation horizon: it constitutes the reference point from which NII variations under stress are measured. Central NII over three years stands at 285.4 M€, i.e., approximately 95 M€ per year, consistent with the net gross margin of 1.25% applied to the 10 Bn€ balance sheet.

Under the parallel upward shock of 200 bp, NII stands at 248.2 M€, a ΔNII of −37.2 M€ (−13.0%). This figure remains well below the EBA alert threshold of 250 M€.[31] Under the −150 bp scenario, NII improves to 313.4 M€ (+27.9 M€), reflecting lower charges on demand deposits.

The semi-annual dynamics under the upward scenario are more instructive than the cumulative figure. NII falls sharply in the first half as liability charges rise before variable asset revenues compensate. It recovers from the second half, when treasury and variable-rate loans, which reprice in less than six months, fully benefit from higher rates. This improvement is however transitory: from year two, NII deteriorates again as term deposits maturing are renewed at higher rates, while long-term fixed-rate assets continue to weigh on the average asset yield.

These results lead to a first observation: NII risk appears manageable. But this conclusion warrants two important qualifications. First, the **regulatory static approach structurally underestimates NII risk**. The EBA's *run-off* assumption supposes that outstanding amounts maturing are not renewed: only the existing stock is measured, frozen in its March 2024 composition. Yet an active bank permanently renews its loans and deposits. Under the constant balance sheet assumption, which supposes that maturing amounts are re-borrowed or re-invested at prevailing market rates, the ΔNII under the upward scenario reaches −237.4 M€ over three cumulative years, versus −37.2 M€ under the static approach: a factor-six difference. Under a constant balance sheet, we are at the limit of the EBA threshold of −250 M€. NII tension is therefore very real in economic terms, even if the standardised regulatory approach does not reveal it. Second, and above all, NII is a short-horizon flow metric that says nothing about the long-term economic value destruction simultaneously occurring over the fifteen to eighteen year residual life of mortgage loans. This is precisely what EVE reveals.

[31] EBA/GL/2022/14, paragraph 109: the NII alert threshold is set at 2.5% of total assets, i.e., $10,000 \times 2.5\% = 250$ M€ for our balance sheet.

Economic value of equity: the true alarm signal

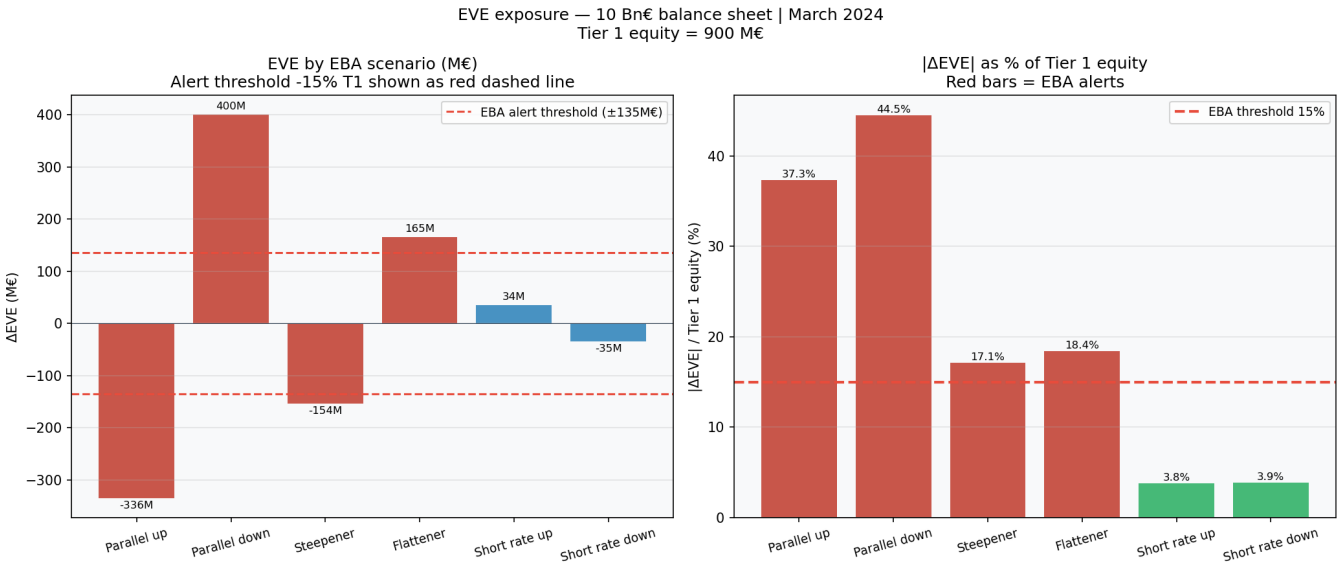

Figure 4: Left: EVE by EBA scenario (M€); the regulatory alert threshold at −15% of Tier 1 is shown as a red dashed line. Right: |ΔEVE| as a percentage of T1 equity, red bars signalling the four scenarios exceeding the 15% threshold.

EVE in the central scenario stands at 1,024.9 M€, slightly above book equity of 900 M€, reflecting the franchise value of commercial relationships and implicit margins. Under the parallel upward shock of 200 bp, EVE falls to 689.1 M€, a ΔEVE of −335.8 M€ representing −37.3% of Tier 1 equity. This result is 2.5 times the 15% regulatory threshold and triggers a prudential alert.

The divergence between ΔNII (−37.2 M€) and ΔEVE (−335.8 M€) under the same 200 bp shock is striking: EVE reveals an exposure nearly ten times greater than that captured by NII. NII only captures the next three years of cash flows. EVE discounts all future cash flows over the residual life of each item. A mortgage extended at 1.80% over eighteen years represents a fixed contractual cash flow whose present value collapses when the discount rate rises from 3.80% to 5.80%. On 3,500 M€ of notional, this discounting effect represents hundreds of millions of economic value loss, even if cash flows are contractually assured.

This mechanism finds its most striking historical illustration in the American savings and loan crisis of the 1980s: these institutions funded long-term fixed-rate mortgages with short-term deposits, exactly as in the modelled balance sheet, and their economic value collapsed when the Fed sharply raised rates under Volcker's impetus. EVE losses, which only materialised in accounting terms upon disposal, remained latent for several years, masking real economic insolvency until the liquidity crisis made them inevitable.[32]

In total, four out of six scenarios trigger the prudential alert: the parallel rise (−37.3% of T1), the parallel fall (+44.5%), the steepener (−17.1%) and the flattener (+18.4%). The IRRBB framework retains the absolute value |ΔEVE| as the surveillance metric: a pronounced rise in EVE under a falling rate scenario reveals the same structural fragility as a loss under a rising scenario. Only the short rate shocks remain compliant (−3.8% and +3.9%), as their impact on the value of long-duration assets is limited.

[32]The savings and loan crisis (1980–1994) led to the failure or resolution of more than 1,000 institutions in the United States, at an estimated cost of 160 billion dollars to the American taxpayer. FDIC, *History of the Eighties, Lessons for the Future*, 1997. https://www.fdic.gov/bank/historical/history/.

Hedging strategy and NII/EVE efficient frontier

Calibrating the hedging portfolio: a bucket-by-bucket approach

The objective of this section is first and foremost to construct a hedging portfolio that satisfies IRRBB regulatory constraints, i.e., bringing $|\Delta\text{EVE}|$ below the 15% of Tier 1 threshold across all six EBA scenarios, while making legible the economic logic underpinning each instrument choice. The efficient frontier of Section 4.2 will then systematically explore all possible coverage levels around this reference portfolio.

The ALM hedging logic consists of matching each instrument to the economic nature of exposed positions, bucket by bucket, rather than posting a global notional uniformly. When rates rise, the received floating flow increases, generating a valuation gain that offsets the loss in economic value of the fixed-rate loan portfolio; symmetrically, a rate fall translates into a loss on swaps that limits the balance sheet's EVE rise. Short *liability-sensitive* buckets reprice naturally with rates and require no explicit hedging; it is on the long *asset-sensitive* buckets, where fixed-rate assets remain frozen at their original rate while funding costs rise, that EVE exposure concentrates. The rate gap calculated in Section 3.1 provides precisely this mapping and the swap portfolio calibration follows directly.

Table 3: Hedging portfolio calibration by maturity bucket

Tranche	Covered buckets	Underlying asset	Net gap	Hedge ratio	Notional	Justification
IRS 3Y	2Y–3Y, 3Y–5Y	Fixed corporate + consumer loans	490 M€	30%	100 M€	Moderate duration, frequent partial renewal
IRS 5Y	5Y–7Y	Sovereign bond portfolio	2,142 M€	60%	1,300 M€	Long duration, concentrated EVE sensitivity
IRS 10Y	7Y–20Y	Fixed-rate mortgage loans	2,936 M€	65%	1,900 M€	Dominant EVE exposure, borrower lock-in
Total					**3,300 M€**	33% of balance sheet

Determination of hedge ratios

The retained *hedge ratios* (30%, 60%, 65%) are not set arbitrarily: they result from a formal optimisation under regulatory constraints, conducted in two steps.

The first step consists of empirically decomposing the total $|\Delta\text{EVE}|$ of 335.8 M€ by tranche (under the parallel upward scenario of 200 bp by tranche, this scenario being the one most directly reflecting the loss in value of long-term fixed-rate assets). Long buckets (7–20 years) bear 57.6% (−334.0 M€), intermediate buckets (5–7 years) 25.7% (−149.0 M€), and short buckets (2–5 years) 15.1% (−87.5 M€). This hierarchy directly underpins the graduated ratios: the instrument whose tranche bears the dominant share of exposure merits the highest ratio.

The second step is a numerical optimisation. The problem is formulated as follows: find the three hedge ratios (h_3, h_5, h_{10}) that minimise the total covered notional (i.e., hedge as *little* as possible) subject to the constraint that $|\Delta\text{EVE}|$ remains below 15% of Tier 1 in each of the six EBA scenarios, both upward and downward:

$$\min_{h_3,h_5,h_{10}} \quad h_3 \cdot G_3 + h_5 \cdot G_5 + h_{10} \cdot G_{10}$$

$$\text{s.t.} \quad |\Delta\text{EVE}(h_3, h_5, h_{10},\, s)| \leq 0.15 \times \text{T1} \quad \forall\, s \in \{6 \text{ EBA scenarios}\}$$

where G_i denotes the net gap of tranche i. This problem is solved numerically using the SLSQP (*Sequential Least Squares Programming*) algorithm.[33] A preliminary sensitivity analysis on the 10Y IRS alone confirms that the

[33] Kraft, D., *A software package for sequential quadratic programming*, Technical Report DFVLR-FB 88-28, DLR German Aerospace Center, 1988.

regulatory compliance zone on the two extreme scenarios is reached between 50% and 80% coverage on this tranche, validating the range of admissible solutions.

The optimal ratios obtained are 30%, 60% and 65%. They are complemented by an economic constraint: short and intermediate buckets are covered at non-zero ratios (30% and 60%) not out of strict regulatory necessity (the 10Y IRS alone would suffice for EVE compliance with the present data) but out of concern for instrument diversification and respect for bucket-by-bucket risk limits of an *ALCO* committee, which reasons line by line rather than solely on consolidated |ΔEVE|.

Short buckets (2–5 years), which carry corporate and consumer loans with short residual maturity and frequent renewal, are covered at 30%. Intermediate buckets (5–7 years), almost exclusively consisting of the sovereign bond portfolio acquired at 1.20%,[34] at 60%. Long buckets (7–20 years), which concentrate almost all fixed-rate mortgages extended between 2015 and 2022 and bear the dominant share of EVE exposure, reach a ratio of 65%.

The resulting portfolio comprises three instruments: a pay-fixed 3Y IRS of 100 M€ at a rate of 3.034%, a pay-fixed 5Y IRS of 1,300 M€ at 2.883%, and a pay-fixed 10Y IRS of 1,900 M€ at 2.810%,[35] for a total notional of 3,300 M€ representing 33% of the balance sheet. The total *DV01* of the portfolio stands at 230.9 k€/bp, meaning that a 1 basis point movement across the entire curve generates a valuation gain or loss of 230,900 € on the swap portfolio.

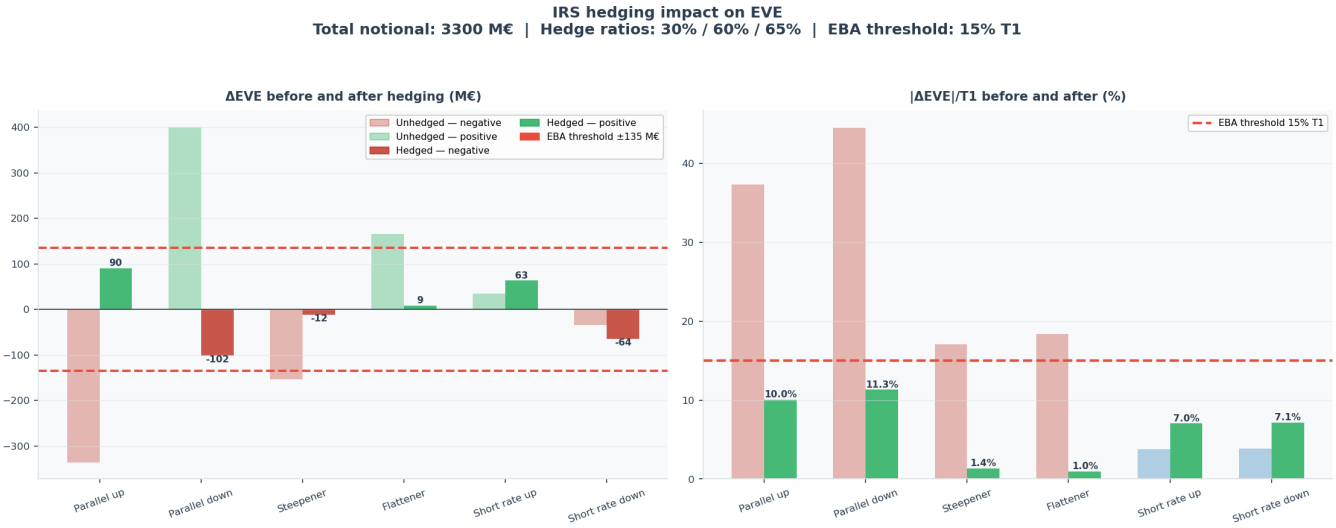

Figure 5: Impact of IRS hedging on EVE. Left: ΔEVE before (transparent bars) and after (solid bars) hedging. Right: |ΔEVE|/T1 before and after hedging; the 15% regulatory threshold is shown as a dashed line.

Results confirm the effectiveness of hedging. Under the parallel upward shock of 200 bp, ΔEVE moves from −335.8 M€ to +90.3 M€ (+10.0% of T1): swaps generate a valuation gain of 426.1 M€ that offsets the value loss on the fixed-rate loan portfolio, while remaining within regulatory limits. Under the parallel downward shock of 200 bp, ΔEVE stands at −102.0 M€ (−11.3% of T1), also compliant. This result contrasts with hedging portfolios calibrated solely on the upward scenario, which mechanically generate over-hedging in the downward scenario: the simultaneous consideration of both extreme scenarios in the optimisation objective function is precisely what avoids this pitfall. In total, all six EBA scenarios are compliant after hedging, versus only two out of six before hedging.

The NII/EVE efficient frontier

The deterministic answer to our research question rests on the construction of the NII/EVE efficient frontier. The principle is as follows: starting from the reference portfolio calibrated in Section 4.1, we vary its total notional from 0 to 150%, which amounts to exploring all possible overall coverage levels while retaining the bucket-by-bucket structure as the backbone. For each level, we calculate the pair formed by the EVE variation and the NII variation under the

[34] Agence France Trésor, *Statistics on government bond issuances*, 2015–2020 data. OAT rates for 5–7 year maturities oscillated between 0.80% and 1.40% over this period. https://www.aft.gouv.fr/fr/encours-detaille-oat.
[35] EUR mid-market swap rates observed on Bloomberg on 29 March 2024. Source: Bloomberg, *EUR IRS Mid Rates*, March 2024.

200 bp upward scenario. The resulting frontier is inspired by Markowitz's concept of the efficient frontier,[36] where NII plays the role of return and EVE that of risk. Compliance is defined here strictly: a point is compliant if and only if $|\Delta\text{EVE}|$ remains below 15% of T1 *both* under the upward and the downward parallel scenarios.

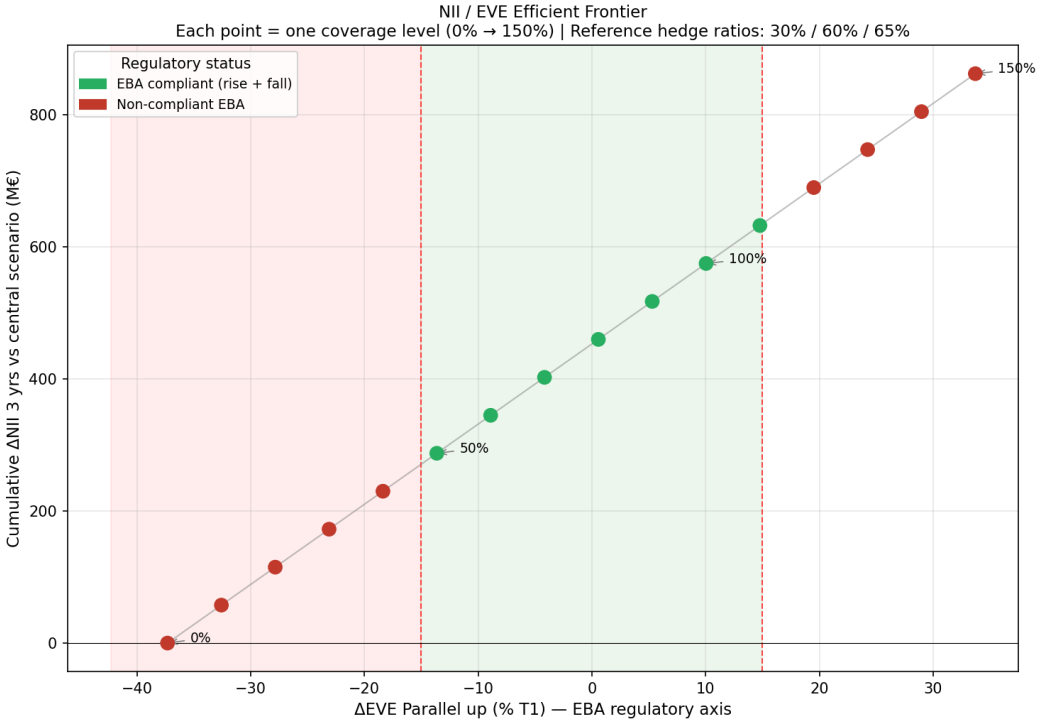

Figure 6: NII/EVE efficient frontier. Each point corresponds to a coverage level (from 0% to 150% of the base portfolio). The green zone corresponds to the region compliant with EBA requirements ($|\Delta\text{EVE}| \leq 15\%$ of T1 in both upward and downward scenarios). Red points are non-compliant. **Reading note**: the ΔNII is here calculated under the upward scenario only; under a falling rate or normalised curve central scenario, this same ΔNII would be negative, as explained in the analysis that follows.

Several lessons emerge from this analysis. The regulatory compliance zone is reached between 50% and 110% of coverage. Within this zone, the cumulative ΔNII under the upward scenario lies between +287.5 M€ and +632.5 M€: hedging therefore simultaneously improves NII and EVE under the upward scenario. This result, counter-intuitive at first glance, is directly explained by the positive carry of March 2024: swaps, in pay-fixed position, receive a high Euribor (3.75%) while paying a lower fixed rate (2.81% for the 10Y IRS), generating a net positive flow added to the balance sheet NII. In other words, hedging generates income in the short term because the curve is inverted. At the reference portfolio (100% coverage), ΔEVE stands at +10.0% of T1 and ΔNII at +575.0 M€ over three years.

It would however be erroneous to interpret this chart as evidence of an arbitrage opportunity, i.e., a situation where the bank would gain in all states of the world simultaneously. What the frontier shows is that under the *sole* upward scenario retained for the vertical axis, the positive carry of swaps causes a positive ΔNII to appear. But this same portfolio generates a net negative flow in a falling rate scenario or in a central scenario with a normalised curve: as Section 4.4 quantifies, a 200 bp ECB rate cut would bring carry to −36.2 M€/year. Hedging does protect EVE in both directions, as is the objective of the optimisation in Section 4.1, but it does not protect NII unconditionally. The NII/EVE dilemma is not permanently resolved: it is *suspended* by the exceptional configuration of March 2024, where the inverted curve makes the regulatory stress scenario coincide with the carry gain scenario.

Beyond 110% of coverage, EVE exposure under the downward scenario exceeds the regulatory alert threshold: optimal coverage therefore lies in the 50–110% zone. In a normally upward-sloping curve configuration, this same zone would exist geometrically, but would only be accessible at the cost of NII compression in the central scenario, where carry becomes negative again and hedging structurally costs on current revenue. This is what Section 4.4 quantifies and what constitutes, in the end, the true dilemma of ALM management in a normalised rate environment.

[36] Markowitz, H., *Portfolio Selection*, *Journal of Finance*, 7(1), 1952, pp. 77–91. https://doi.org/10.1111/j.1540-6261.1952.tb01525.x.

Carry cost and the macroeconomic window: the cornerstone of compatibility

We have just shown that the NII/EVE compliance zone is entirely positive on the NII axis under the upward scenario. This section explains the underlying reason and measures its transitory nature: it is the swap carry, made positive by the curve inversion, that is the cornerstone of this compatibility.

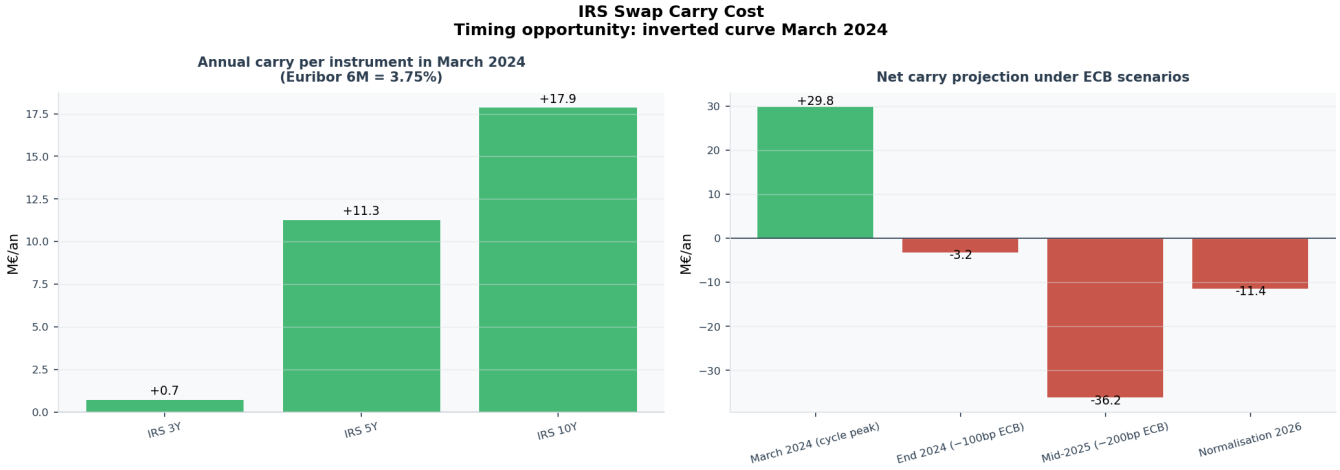

Figure 7: IRS swap carry cost. Left: annual carry per instrument in March 2024 (Euribor 6M = 3.75%). Right: projection of net carry under anticipated ECB rate evolution scenarios.

The 3,300 M€ swap portfolio generates a net positive carry of +29.8 M€/year in March 2024: the 3Y IRS contributes +0.7 M€, the 5Y IRS +11.3 M€, and the 10Y IRS +17.9 M€. Positive carry here corresponds to the difference between the received floating rate (Euribor 6M at 3.75%) and the paid fixed rate (2.81% for the 10Y IRS, for example): since short rates exceed long rates in an inverted curve configuration, the bank receives more than it pays on the floating flow, generating net revenue rather than a cost. Whereas the same instruments would have cost 30 to 40 M€/year in a normal curve configuration, they now generate this net revenue. This reversal is what operationally enables the NII/EVE compatibility evidenced by the efficient frontier.

This window is however transitory and its closure is mechanical. If the ECB cuts rates by 100 bp by end-2024, Euribor would fall to 2.75% and carry would shift to −3.2 M€/year. A 200 bp cut, anticipated by mid-2025, would bring it to −36.2 M€/year. At the horizon of full normalisation (Euribor at 2.50% in 2026), annual carry would stand at −11.4 M€/year.[37]

This result directly conditions the answer to the central research question. NII/EVE compatibility is not a permanent state: it is a window whose existence depends jointly on the level of coverage and the macro-financial configuration. March 2024 is precisely the moment when hedging is both a regulatory necessity and economically advantageous, and this coincidence will not last.

Sensitivity to behavioural assumptions

The retained behavioural assumptions (three-year demand deposit duration, 5% prepayment floor rate) are estimation parameters. A sensitivity analysis over nine combinations (demand deposit duration of 2, 3 or 4 years crossed with a prepayment rate of 3%, 5% or 8%) reveals the robustness of the central conclusion.

[37] These ECB rate evolution scenarios are constructed from March 2024 market expectations: Euribor futures (3M Euribor futures, December 2024 and June 2025 expiries) priced in at that date a cumulative cut of 100 to 150 bp by end-2024 and 200 bp by mid-2025. Source: Bloomberg, *Euribor futures curve*, 29 March 2024. The normalisation scenario at 2.50% is consistent with ECB median projections for the euro area long-term neutral rate. ECB, *Survey of Professional Forecasters*, first quarter 2024. https://www.ecb.europa.eu/stats/ecb_surveys/survey_of_professional_forecasters/html/index.en.html.

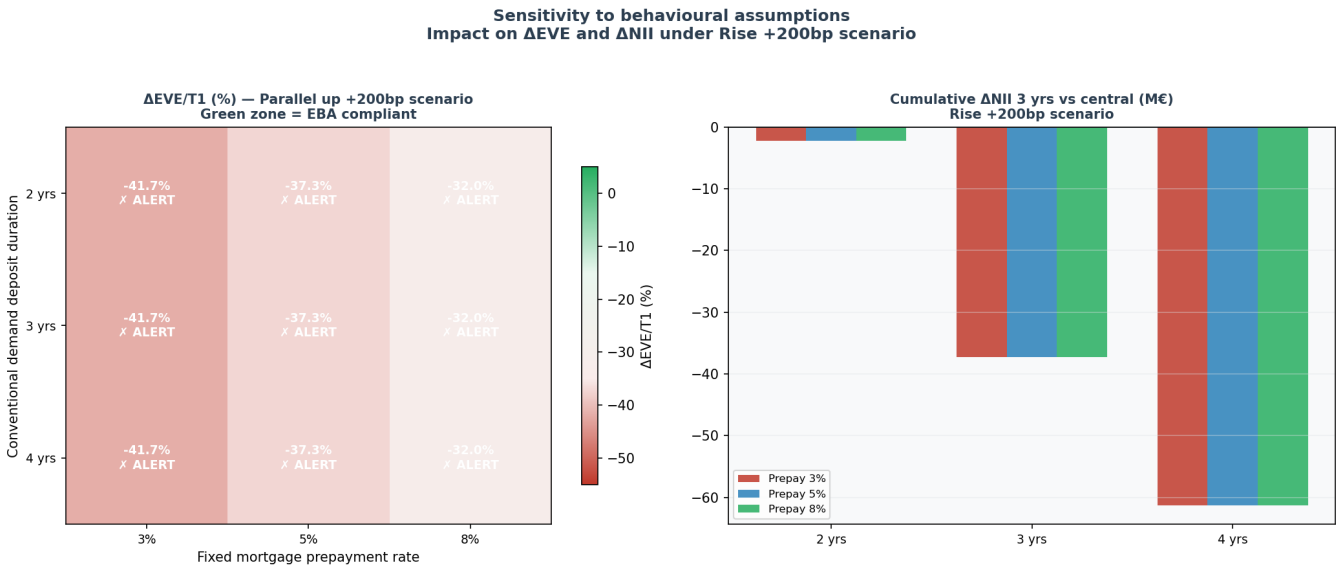

Figure 8: Sensitivity to behavioural assumptions. Left: heatmap of ΔEVE/T1 under the Parallel up +200 bp scenario for the nine combinations tested. Right: cumulative ΔNII over three years under the upward scenario for the same assumptions.

Results are unambiguous: across the nine tested combinations, **none is compliant with EBA thresholds without hedging**. The central assumption (3Y demand deposits, 5% prepayment) yields a ΔEVE of −37.3% of T1; the most optimistic assumption (4Y demand deposits, 8% prepayment) yields −32.0%: still more than double the regulatory threshold. The necessity to hedge is therefore robust across the entire plausible parameterisation spectrum.

We note the dissociation between EVE and NII sensitivity to behavioural parameters: ΔEVE varies from −41.7% (2Y demand deposits) to −32.0% (4Y demand deposits, 8% prepayment), a 9.7-point spread, while ΔNII ranges from −2.2 M€ (2Y demand deposits) to −61.3 M€ (4Y demand deposits). This illustrates that demand deposit duration affects NII more than EVE in this balance sheet, and that prepayment has no impact on NII in the static approach. This result anticipates the conclusion of Extension B: demand deposit duration is indeed the most influential behavioural parameter.

Extensions: towards a stochastic efficient frontier

The deterministic efficient frontier of Section 4.2 answers the research question, but it rests on two simplifying assumptions whose robustness must be tested. On one hand, it is constructed for a single rate trajectory: it tells us what happens if rates rise by 200 bp, but says nothing about the probability that this compliance holds across all plausible trajectories. On the other hand, client behaviours, prepayment and deposit mobility, are frozen at their March 2024 values, whereas they in practice adapt to the rate level. The two extensions that follow successively lift these limitations.

From binary to probabilistic compliance: Monte Carlo simulation using the Hull-White model

Principle and model justification

The objective of this extension is to measure, not whether the hedging portfolio is compliant under the sole parallel upward shock of 200 bp retained in Section 4.2 (under a fixed rate scenario), but with what probability it remains compliant across all plausible rate trajectories drawn from the distribution calibrated on March 2024.

We replace the single deterministic scenario with a distribution of 1,000 simulated rate trajectories. The retained model is the single-factor Hull-White model.[38] It generates stochastic trajectories of the short rate whose mean exactly reproduces the EUR OIS curve of March 2024, thereby ensuring that simulations remain anchored in the macro-financial reality that constitutes the central framework of our analysis. Its fundamental equation describes how the short rate evolves at each instant:

$$dr(t) = \big[\theta(t) - \kappa\, r(t)\big]\, dt + \sigma\, dW(t)$$

The three terms have a direct economic interpretation. The term $-\kappa\, r(t)$ is a restoring force: if the short rate rises too high, it is pulled back down, and vice versa. This is the *mean-reversion*, characteristic of interest rates. The term $\sigma\, dW(t)$ introduces randomness: at each instant, an unpredictable shock of size σ perturbs the trajectory. The variable $\theta(t)$ is a deterministic function of time, analytically calibrated so that the mean of trajectories exactly reproduces the EUR OIS curve observed in March 2024:

$$\theta(t) = \frac{\partial f}{\partial t}(0,t) + \kappa\, f(0,t) + \frac{\sigma^2}{2\kappa}\big(1 - e^{-2\kappa t}\big)$$

where $f(0,t)$ is the instantaneous forward rate extracted from the zero-coupon curve by differentiating the cubic spline. This property guarantees $\mathbb{E}[r(t)] = f(0,t)$: in expectation, trajectories start exactly from the market curve, verified by comparing the mean of the 1,000 simulated trajectories with observed forward rates (deviations below $\pm 0.15\%$ across all horizons).

This is the fundamental difference with the Vasicek model, whose constant drift θ cannot reproduce an inverted curve of 107 bp between the three-month and ten-year rates. Using Vasicek here would amount to simulating trajectories whose expected starting point would structurally diverge from the central curve of our analysis, invalidating the link between simulation and the March 2024 macro-financial reality.

The retained parameters are: $r(0) = 3.85\%$ (3M Euribor observed in March 2024[39]), $\kappa = 0.10$ (mean-reversion speed calibrated on historical ECB monetary policy cycles over the 2000–2024 period; this value implies a half-life of the rate shock of approximately seven years, consistent with empirical estimates by Brigo and Mercurio on European short rates[40]) and $\sigma = 1.00\%$ (calibrated on normal ATM EUR swaption volatilities published by the ECB, in the order of 90 bp/year over the 2023–2024 period[41]). The model is Gaussian, so negative rates are theoretically possible;

[38] Hull, J., White, A., *Pricing interest-rate-derivative securities*, *Review of Financial Studies*, 3(4), 1990, pp. 573–592. https://doi.org/10.1093/rfs/3.4.573.
[39] ECB, *Euro interbank offered rates (EURIBOR)*, historical data, March 2024.
[40] Brigo, D., Mercurio, F., *Interest Rate Models — Theory and Practice*, 2nd edition, Springer Finance, 2006, pp. 73–75.
[41] ECB, *Euro area yield curves and interest rate statistics*, implied volatility data, March 2024. https://www.ecb.europa.eu/stats/financial_markets_and_interest_rates/euro_area_yield_curves/html/index.en.html.

in practice, starting from $r_0 = 3.85\%$ with $\sigma = 1\%$ over three years, no trajectory entered negative territory in our simulation.

Simulation protocol

Trajectories are discretised using an *Euler-Maruyama* scheme,[42] with a quarterly time step $dt = 0.25$ yr, i.e., the short rate is calculated each quarter by applying the following formula:

$$r(t_{i+1}) = r(t_i) + \big[\theta(t_i) - \kappa\, r(t_i)\big]\, dt + \sigma\sqrt{dt}\, \varepsilon_i, \quad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$$

The first term on the right-hand side is the short rate in the previous quarter. The second term is the deterministic drift, pulling the rate towards its equilibrium value. The third term is the quarterly random shock. A floor of 0% is applied for consistency with post-2014 ECB rate policy.

For each trajectory i, we calculate the mean short rate over the first year $\bar{r}_i^{(1)}$, applied as a parallel shift of the base curve. The NII and EVE calculation functions from the central model are then called for each trajectory, producing a cloud of 1,000 pairs (ΔEVE/T1, ΔNII).

Results

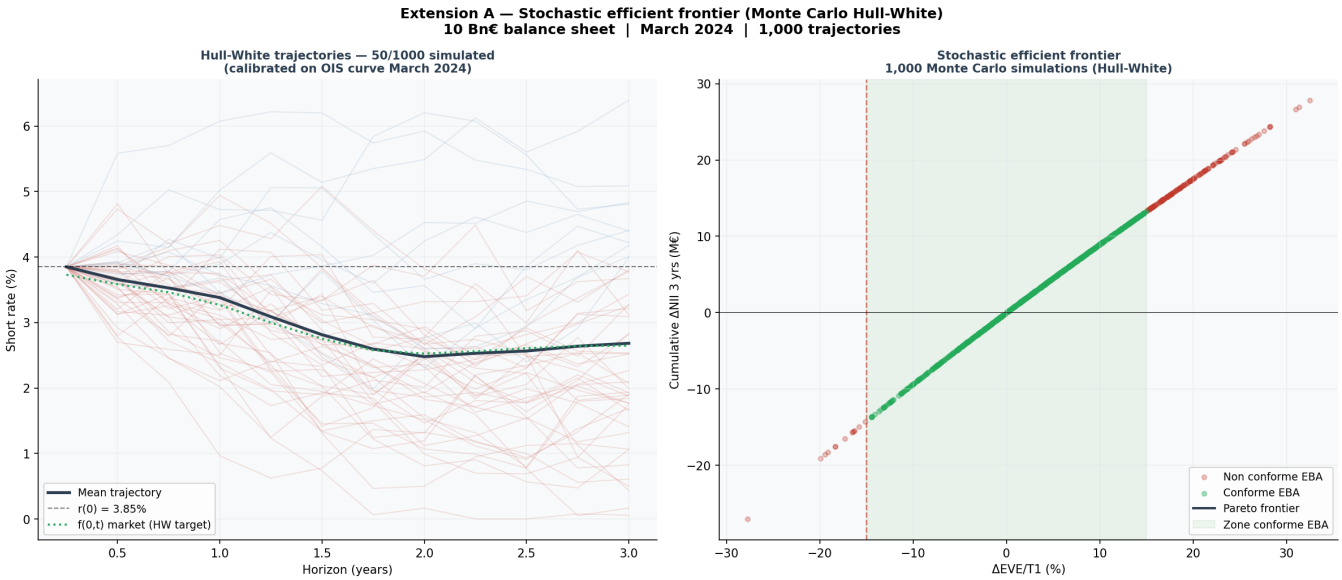

Figure 9: Stochastic efficient frontier (Hull-White Monte Carlo). Left: 50 out of 1,000 simulated trajectories, with the mean trajectory and the market forward $f(0,t)$ as calibration target. Right: cloud of 1,000 pairs (ΔEVE/T1, ΔNII), distinguishing EBA-compliant/non-compliant and the Pareto frontier.

Out of 1,000 trajectories, **833 are compliant with EBA thresholds (83.3%)**. Median ΔNII stands at +4.7 M€, median ΔEVE at +5.2% of T1, and ΔEVE at the 95th percentile (most adverse scenario in the distribution) at −10.4% of T1, below the 15% regulatory threshold.

Three lessons emerge. First, compliance is likely but not certain: the deterministic frontier identified a compliance zone for 50 to 110% coverage, but the Monte Carlo specifies that this zone is accessible with a probability of 83.3% under the calibrated distribution. The 16.7% of non-compliant trajectories correspond primarily to scenarios where rates remain elevated for longer than anticipated. Second, the positive median ΔNII confirms the argument of Section 4.4: under the central distribution of trajectories, predominantly oriented towards a gradual decline to 2.5%, swap carry improves net NII. Third, the ΔEVE at the 95th percentile of −10.4% shows that the hedging portfolio effectively protects even the distribution tails: in the 5% most adverse trajectories, the prudential alert is not triggered.

[42] Kloeden, P.E., Platen, E., *Numerical Solution of Stochastic Differential Equations*, Springer, 1992, pp. 340–341. The Euler-Maruyama scheme is the stochastic generalisation of the classical Euler scheme: it converges in mean square to the exact solution with a convergence order of 0.5 in time step, which is sufficient for our application given the quarterly time step retained.

The *Pareto* frontier further reveals that, at a given ΔEVE, ΔNII exhibits significant variance across trajectories. This illustrates a point that the deterministic frontier could not capture: it is not only the average shock level that determines NII, but the shape of the trajectory, the speed of the decline, any rebounds, that modulates current revenue quarter by quarter.

Endogenous client behaviours and compliance robustness: a dynamic Monte Carlo loop

Motivation

The previous extension enriched the stochastic dimension of the frontier, but continued to freeze client behaviours at their March 2024 values. Yet these behaviours are endogenous to the rate level: a borrower refinances their loan when market rates fall below their contractual rate, and a depositor moves towards higher-yielding products when short rates rise. This extension integrates these two feedbacks into the Monte Carlo loop, over 500 Hull-White trajectories, to test whether the compliance zone identified previously is robust when the balance sheet adapts to the rate trajectory.

Dynamic behavioural functions

The **prepayment rate** is modelled by a *logistic sigmoid* of the incentive to refinance:

$$\text{prepay}(r_{\text{market}}, r_{\text{contract}}) = p_{\min} + (p_{\max} - p_{\min}) \cdot \frac{1}{1 + e^{-\alpha\,(r_{\text{contract}} - r_{\text{market}})}}$$

with $p_{\min} = 5\%$ (lock-in floor observed in March 2024), $p_{\max} = 22\%$[43] and $\alpha = 2.5$.[44] The logic is simple: when the market rate exceeds the contractual rate, as in March 2024 where the spread reaches +240 bp, no one refinances; when it falls below, the incentive rises rapidly.

The **conventional duration of demand deposits** is a linearly decreasing function of the short rate:

$$\text{demand_deposit_duration}(r_{\text{short}}) = \max\Big(1.0\ ;\ \min\Big(4.5\ ;\ 3.0 - 0.40 \times \max(0\ ;\ r_{\text{short}} - 1.5\%)\Big)\Big)$$

Each percentage point of short rate above 1.5% reduces demand deposit duration by 0.40 years, capturing the arbitrage of savers towards savings accounts or term deposits when the return on risk-free investments becomes attractive. The coefficient $\beta = 0.40$ and the 1.5% threshold are calibrated on the empirical estimates of Memmel (2011),[45] with bounds of 1 and 4.5 years consistent with the EBA regulatory cap of 5 years and French market practice.

[43] Banque de France, *Stat Info, Production de crédits immobiliers*, 2016–2017 data. Prepayment rates reached 15 to 20% during the 2016–2017 refinancing waves, versus 5 to 7% in lock-in periods. https://www.banque-france.fr/statistiques/taux-et-cours/taux-des-credits.

[44] The logistic form has been standard in the prepayment literature since Richard and Roll (1989). The value $\alpha = 2.5$ is calibrated to reproduce the transition speed observed between lock-in and active refinancing during the 2014–2016 rate cut cycle: a 100 bp spread between contractual and market rates brings prepayment from its floor to approximately 12%, consistent with French data from that period. Richard, S.F., Roll, R., *Prepayments on fixed-rate mortgage-backed securities*, *Journal of Portfolio Management*, 15(3), 1989, pp. 73–82.

[45] Memmel, C., *Banks' exposure to interest rate risk*, Deutsche Bundesbank, Discussion Paper 07/2011, pp. 12–14. The author documents a significant reduction in the effective duration of demand deposits in German institutions during episodes of short rate rises above 1.5%. https://doi.org/10.1016/j.jbankfin.2010.08.003.

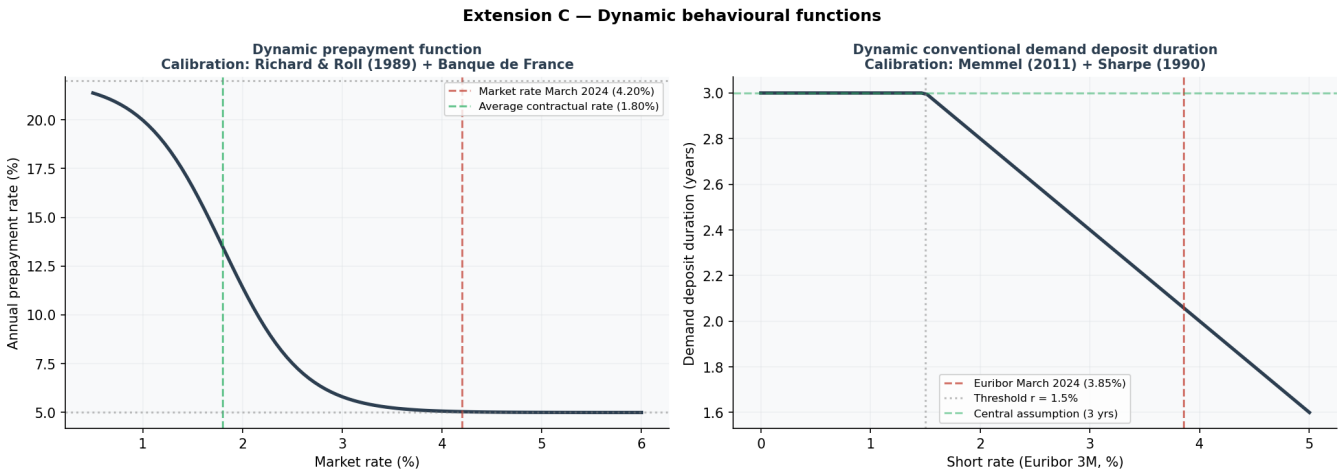

Figure 10: Dynamic behavioural functions. Left: logistic prepayment rate as a function of the market rate (contractual rate at 1.80%). Right: conventional demand deposit duration as a function of the short rate. Dashed lines indicate values retained in March 2024.

Results

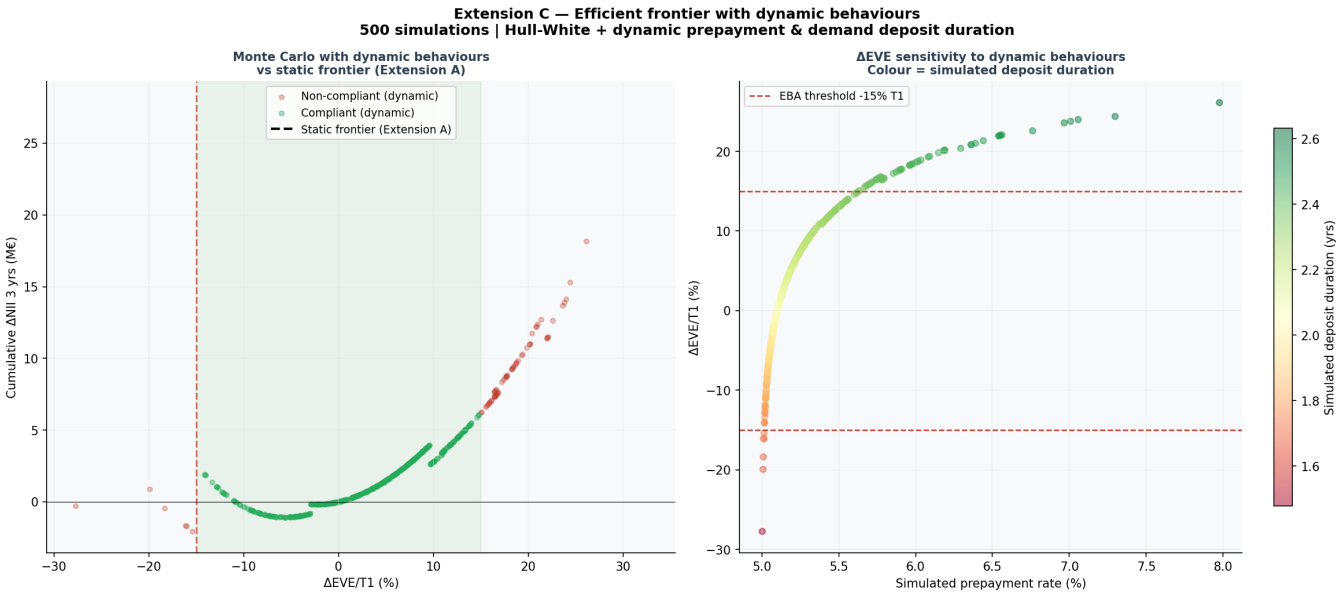

Figure 11: Monte Carlo with dynamic behaviours (500 trajectories). Left: (ΔEVE/T1, ΔNII) cloud with dynamic behaviours superimposed on the static frontier from the previous extension. Right: ΔEVE sensitivity to simulated prepayment, coloured by simulated demand deposit duration.

Out of 500 trajectories, **420 are compliant with EBA thresholds (84.0%)**, slightly higher than in the previous extension (83.3%). This result warrants explanation, as it is counter-intuitive: making behaviours endogenous improves EVE compliance rather than degrading it.

The mechanism is as follows. Under the calibrated Hull-White distribution, the short rate remains on average above 1.5% over virtually the entire horizon, activating the reduction in demand deposit duration: mean simulated duration is 2.12 years versus 3 years in the static model. A shorter demand deposit duration shortens the conventional maturity of liabilities, which reduces the asset-liability maturity gap and therefore EVE exposure under an upward shock. In other words, depositors who arbitrage more rapidly towards other products paradoxically contribute to reducing the structural interest rate risk of the balance sheet, by bringing liability duration closer to that of short assets.

The counterpart is NII deterioration: a liability repricing more rapidly sees its interest charges rise faster under an upward shock. Median ΔNII recedes from +4.7 M€ (previous extension) to +1.1 M€: EVE compliance is preserved, but NII headroom shrinks.

This result extends and validates the static sensitivity analysis of Section 4.3: demand deposit duration is the most influential behavioural parameter on EVE, and its endogenous shortening in a high-rate environment is simultaneously an NII risk and an EVE resilience factor, illustrating the complexity of interactions between the two ALM metrics when the balance sheet adapts to the rate trajectory.

Conclusion

Answer to the research question

The 2022–2024 rate shock did not render IRRBB compliance and NII preservation incompatible. But this conclusion, stated without nuance, obscures the dynamics that produce it and the fragile conditions under which it holds.

The more accurate reading is as follows. The shock did not create a new incompatibility between the two regulatory metrics; it revealed a structural fragility silently accumulated over a decade of low rates. The NII/EVE tension existed before 2022, but lay dormant: regulatory EVE variations remained modest as long as rate shocks were applied to near-zero levels, and the prohibitive cost of hedging, up to 30–40 M€/year of negative carry, rendered any preventive action economically irrational. The 2022–2024 shock simultaneously activated the EVE constraint, pushing $|\Delta\text{EVE}|$ to 2.5 times the regulatory threshold, and, paradoxically, reduced the cost of hedging, by inverting the curve and making swap carry positive at +29.8 M€/year. This exceptional coincidence between regulatory necessity and economic opportunity is precisely what makes NII/EVE compatibility accessible in March 2024 within a coverage zone between 50% and 110% of the calibrated swap portfolio.

The calibration of hedge ratios furthermore illustrates how a rigorous approach can transcend a simple reasoned management decision: the ratios of 30%, 60% and 65% result from a formal optimisation that minimises total covered notional subject to EBA compliance across all six regulatory scenarios, both upward and downward. This dual constraint is precisely what avoids over-hedging under the downward scenario, a classic pitfall of strategies calibrated on a single upward scenario. The result confirms that the 10Y IRS is the dominant lever: long buckets bear 57.6% of total $|\Delta\text{EVE}|$, justifying the highest ratio on this tranche. The resulting portfolio of 3,300 M€ (33% of the balance sheet) brings all six EBA scenarios below the 15% Tier 1 threshold, with $|\Delta\text{EVE}|$ of 10.0% under the upward and 11.3% under the downward parallel scenarios.

The deterministic efficient frontier establishes this compatibility for a single rate scenario. The Monte Carlo extensions refine and nuance it: it is accessible with a probability of 83 to 84% under the distribution of plausible trajectories calibrated on the March 2024 OIS curve, but 16 to 17% of trajectories, primarily those where rates remain elevated for longer than anticipated, keep the balance sheet in prudential alert despite hedging. The integration of endogenous client behaviours further reveals a counter-intuitive mechanism: the shortening of demand deposit duration, endogenous to the rate level, paradoxically improves EVE compliance while slightly compressing NII, illustrating that the two metrics respond differently to the same behavioural levers.

The structurally robust conclusion is therefore as follows: there is no fundamental incompatibility between IRRBB compliance and NII preservation, but operational compatibility is probabilistic, narrow, and conditional on the macro-financial configuration. It will no longer be accessible without cost once rate normalisation is complete: once the ECB has brought rates back to a neutral level, swap carry will turn negative again, and the compliance zone of the efficient frontier will only exist at the cost of sacrificing NII in the central scenario.

Perspectives and limitations

This paper rests on a number of acknowledged simplifications whose implications for future work deserve to be made explicit.

The first limitation relates to the **run-off approach for NII**. The EBA regulatory framework requires that maturing amounts not be renewed, which structurally underestimates NII risk for an active bank. We have shown that ΔNII moves from -37.2 M€ under the static approach to -237.4 M€ under a constant balance sheet, a factor of six: NII tension is therefore very real in economic terms, even if the standardised regulatory approach does not fully reveal it. A natural enrichment would consist of integrating realistic new production assumptions into the dynamic NII calculation, in line with the developments envisaged by the EBA in its forthcoming IRRBB guideline revisions.

The second limitation concerns **single-factor modelling**. The single-factor Hull-White model describes short rate

dynamics through a single source of randomness, implying that the entire curve shifts parallel in expectation. This approximation, classical in ALM for short-horizon simulations, underestimates slope and curvature effects that affect different curve maturities differently. By construction, it also tends to underestimate $|\Delta\text{EVE}|$ in steepening trajectories, which means the 83.3% compliance rate obtained in Extension A is slightly optimistic. A two-factor model, or a full rate model such as the LMM (*Libor Market Model*), would capture these shocks endogenously.

The third limitation concerns **instrument convexity** and implicit optionality. The *DV01* calculation assumes a linear relationship between rate variation and value variation. For long-duration mortgages with prepayment options, negative convexity under a large upward shock is a non-negligible factor that our model does not explicitly integrate. Similarly, prepayment options constitute implicit optionality in the balance sheet whose valuation would require a rate model with non-parallel curve deformation.

The fourth limitation is the **static modelling of liability structure**. Extension B's behavioural functions make demand deposit duration and prepayment endogenous to the rate trajectory, but they do not model substitutions between liability categories, notably depositor arbitrage from demand deposits towards savings accounts or term deposits, nor competitive pricing dynamics. In practice, an ALCO committee's funding decisions integrate liquidity and marginal cost of resources constraints that interact with interest rate risk management in ways that our stylised model does not capture.

The fifth limitation concerns **hedge ratio optimisation**. The SLSQP algorithm used is a local non-linear programming solver: it guarantees convergence to a local but not necessarily global optimum if the objective surface is non-convex. Validation by exhaustive grid confirms the consistency of the retained solution (30%, 60%, 65%), but a genetic algorithm or stochastic global optimisation approach would provide a more robust guarantee. Furthermore, the retained objective, minimising total covered notional, is an approximation of the true ALCO decision problem, which would also reason in terms of carry cost, instrument liquidity and DV01 limit constraints.

Beyond limitations, this paper highlights a structural challenge for the ALM management of French retail banks in the years ahead. The stock of low-rate mortgages will remain on the balance sheet for another ten to fifteen years, maintaining pressure on EVE even as new production benefits from more favourable conditions. The favourable hedging window that March 2024 offered is now closing, as the ECB has effectively begun its rate cut cycle: the net carry on the swap portfolio, still positive at +29.8 M€/year in March 2024, turned negative by end-2024. Institutions that did not seize this opportunity will have to manage NII/EVE tension in a normally upward-sloping curve environment, where hedging has a direct cost on current revenue and where the compliance zone of the efficient frontier is no longer free.

From a methodological standpoint, the NII/EVE efficient frontier we have constructed constitutes, beyond its academic interest, a genuine decision-making tool for an ALCO committee. By enabling visualisation of all accessible combinations, identification of the compliance zone, and measurement of its probability under different market and client behaviour assumptions, it transforms a multidimensional optimisation problem into a legible decision for institutional governance. This is, in the end, what one expects of a good ALM model: not to eliminate uncertainty, but to make it governable.

Glossary

The following terms are defined in alphabetical order. Each occurrence in the text is signalled in italics and refers back to this section.

ALCO (Asset and Liability Committee) — A bank's asset-liability management committee, responsible for supervising interest rate, liquidity and foreign exchange risk in the banking book. It sets risk limits, validates hedging strategies and arbitrates between NII and EVE metrics.

ALM (Asset and Liability Management) — The set of management techniques aimed at controlling financial risks arising from the asymmetry between a bank's assets and liabilities, in particular interest rate risk, liquidity risk and foreign exchange risk in the banking book.

Asset-sensitive (asset-sensitive position) — Describes a balance sheet or maturity tranche whose assets reprice faster than liabilities. An asset-sensitive position benefits from rising rates (interest revenues increase before charges) and suffers from falling rates. Opposed to *liability-sensitive*.

Banking book — The set of positions held by the bank for the purposes of managing its commercial activity and not for trading: customer loans, deposits, investment bond portfolios. It is opposed to the *trading book*, which contains positions held for short-term transaction purposes.

Maturity bucket — A standardised time interval grouping asset and liability flows due for repricing within the same window, according to the EBA breakdown: 0–3 months, 3–6 months, 6 months–1 year, 1–2 years, 2–3 years, 3–5 years, 5–7 years, 7–10 years, 10–15 years, 15–20 years, over 20 years.

Carry (carry cost) — The periodic net cash flow generated by a hedging instrument, here the difference between the received floating rate (Euribor) and the paid fixed rate on an IRS. In a normally upward-sloping curve configuration, the carry on a pay-fixed IRS is negative (the bank pays more than it receives). In an inverted curve, as in March 2024, carry becomes positive: the received floating leg (3.75%) exceeds the paid fixed leg (2.81% for the 10Y IRS).

Parallel shock — A simultaneous and uniform shift of all points on the yield curve by the same number of basis points, either upward or downward. Opposed to non-parallel shocks (steepening, flattening, short rate shocks) which affect different maturities differently.

Convexity — A measure of the curvature of the relationship between an instrument's value and interest rates. A positively convex instrument gains more in value than it loses for symmetric rate variations (classic bonds). A negatively convex instrument (such as a pay-fixed IRS calibrated on the upward scenario) loses more in the opposite scenario than it gains in the target scenario.

Demand deposits — Bank deposits repayable at any time at the customer's request. Although contractually of zero duration, they exhibit a significantly longer economic duration due to switching costs and loyalty effects. The EBA imposes a conventional duration capped at five years for their integration in IRRBB calculations.

Duration — A measure of the sensitivity of a financial instrument's value to a change in interest rates, or equivalently, the weighted average maturity of its discounted future cash flows. A fixed-rate twenty-year mortgage has a duration close to twelve to fourteen years. The higher the duration, the more sensitive the instrument is to rate variations.

DV01 (Dollar Value of a Basis Point) — The mark-to-market value change of an instrument for a 1 basis point (0.01%) shift in the yield curve. The reference sensitivity metric in bond portfolio and rate hedging management. The DV01 of this paper's swap portfolio stands at 281.2 k€/bp.

Euler-Maruyama scheme — A numerical method for discretising stochastic differential equations, consisting of approximating the continuous solution by a sequence of discrete steps combining the deterministic drift and the random shock of the model. Used here to simulate short rate trajectories of the Hull-White model on a quarterly time step.

NII/EVE efficient frontier — The set of (NII variation, EVE variation) pairs accessible by varying the level of interest rate swap coverage, for a given balance sheet portfolio under a stress scenario. Adapted from Markowitz's (1952) efficient frontier concept to the ALM framework: NII plays the role of return and EVE that of risk.

Rate gap — For each maturity bucket, the difference between outstanding assets and liabilities liable to reprice within that period. A negative gap (liabilities > assets) signals a *liability-sensitive* position; a positive gap signals an *asset-sensitive* position.

Hedge ratio — The proportion of a nominal exposure covered by a hedging instrument. A hedge ratio of 80% on mortgage loans means that 80% of the asset-sensitive gap identified on that maturity tranche is covered by a corresponding notional IRS.

IFRS 9 (International Financial Reporting Standard 9) — The international accounting standard governing the classification, measurement and recognition of financial instruments. Regarding hedging, IFRS 9 requires derivatives to be measured at fair value (*mark-to-market*) and governs the conditions for hedge accounting.

IRS (Interest Rate Swap) — A contract whereby two counterparties periodically exchange interest flows calculated on a reference notional: one pays a fixed rate agreed at inception (fixed leg) and receives a variable rate indexed on Euribor (floating leg). A pay-fixed IRS is used to hedge an upward rate exposure: if rates rise, the received floating leg increases, generating a gain that offsets the balance sheet's economic value loss.

Liability-sensitive (liability-sensitive position) — Describes a balance sheet whose liabilities reprice faster than assets. A *liability-sensitive* position suffers from rising rates (interest charges increase before revenues) and benefits from falling rates. This is the structural position of French retail banks heavily exposed to fixed-rate mortgages.

Lock-in effect — A situation in which a borrower holding a low fixed-rate loan has no incentive to refinance or prepay, because the current market rate exceeds their contractual rate. In March 2024, with an average contractual rate of 1.80% versus a market rate of 4.20%, the lock-in effect is maximal and prepayments are reduced to constrained components (relocation, death, divorce).

Net interest margin — The spread between the average rate on interest-bearing assets and the average rate on remunerated liabilities. It constitutes the primary source of revenue for retail banks. In this paper, the net gross margin of the stylised balance sheet stands at 1.25% (2.73% on the asset side versus 1.48% on the liability side).

Transformation margin — The component of net interest income corresponding to the gain realised from the spread between the return on long assets and the cost of short liabilities. It constitutes the economic remuneration of the interest rate risk borne by the bank in its maturity transformation activity: the greater the duration gap between assets and liabilities, the greater the potential margin and the greater the interest rate risk exposure.

Mark-to-market (MTM) — Valuation of a financial instrument at its instantaneous market value, as opposed to historical cost or amortised cost. For an IRS, MTM is the discounted value of all future cash flows (fixed leg minus floating leg) at current market rates. A zero initial MTM means the instrument is put in place at balanced market conditions.

Pareto frontier — The set of non-dominated solutions to a multi-objective optimisation problem. A point is on the Pareto frontier if no other solution simultaneously improves all objectives. In the Monte Carlo cloud context, a pair $(\Delta\text{EVE}, \Delta\text{NII})$ is on the frontier if no other trajectory simultaneously presents a more favourable ΔEVE and a higher ΔNII.

Pillar 3 (market discipline) — The third pillar of the Basel III prudential framework, dedicated to transparency. It requires credit institutions to publish quantitative and qualitative information on their risk profile, including their IRRBB exposure, EVE variations under regulatory stress scenarios and their client behaviour modelling methods.

Prepayment — The repayment of a loan before its contractual maturity. In France, any borrower may prepay their mortgage subject to early repayment penalties capped by law. The prepayment rate aggregates voluntary repayments (refinancing, property sale) and constrained repayments (death, divorce).

Repricing — The revision of a financial instrument's rate to prevailing market conditions, either at maturity (for fixed-rate instruments) or periodically (for variable-rate instruments). Repricing of a demand deposit is nearly instantaneous; that of an eighteen-year fixed-rate mortgage only occurs at contract maturity.

Mean-reversion — A property of rate models whereby the short rate is attracted towards a long-term equilibrium level when it deviates from it. In the Hull-White model, the parameter κ controls the speed of this reversion: a value of 0.10 corresponds to mean-reversion over approximately ten years.

Logistic sigmoid — An S-shaped mathematical function with values in $[0, 1]$, used to model probabilities or transition rates between two extreme states. In this paper, it describes the transition of prepayment between its lock-in floor (5%) and its active refinancing ceiling (22%) as a function of the economic incentive to refinance.

Tier 1 capital — The core layer of a bank's regulatory capital, consisting essentially of paid-in share capital, reserves and retained earnings. In this paper, Tier 1 capital stands at 900 M€. The EBA alert threshold on EVE is expressed as a percentage of Tier 1: a $|\Delta\text{EVE}|$ exceeding 15% of Tier 1 signals that a rate shock would destroy more than one seventh of the institution's loss-absorption capacity.

Maturity transformation — The founding activity of retail banking, consisting of collecting short-term resources (demand deposits, savings accounts) and deploying them in long-term assets (mortgage loans, sovereign bonds). This duration asymmetry is the primary source of bank net interest income, but it exposes the institution to interest rate risk in the banking book.

References

[1] **Adam, A.** (2007). *Handbook of Asset and Liability Management: From Models to Optimal Return Strategies.* Wiley Finance.

[2] **Banque de France** (2024). *Stat Info, Production de crédits immobiliers, données 2022–2023.* Direction des statistiques monétaires et financières. https://www.banque-france.fr/statistiques/taux-et-cours/taux-des-credits

[3] **BCBS** (2016). *Interest rate risk in the banking book.* Standards, April 2016. Basel Committee on Banking Supervision. https://www.bis.org/bcbs/publ/d368.htm

[4] **ECB** (2024). *Key ECB interest rates, historical data.* Statistical Data Warehouse, March 2024. https://www.ecb.europa.eu/stats/policy_and_exchange_rates/key_ecb_interest_rates/html/index.en.html

[5] **BNP Paribas** (2024). *Rapport sur les risques, Pilier 3, exercice 2023.* IRRBB sensitivity table, p. 214. https://invest.bnpparibas/document/document-denregistrement-universel-et-rapport-financier-annuel-2023-pdf

[6] **Campbell, J.Y., Shiller, R.J.** (1991). Yield spreads and interest rate movements: A bird's eye view. *Review of Economic Studies*, 58(3), 495–514. https://doi.org/10.2307/2298008

[7] **Culbertson, J.M.** (1957). The term structure of interest rates. *Quarterly Journal of Economics*, 71(4), 485–517.

[8] **EBA** (2022). *Guidelines on the management of interest rate risk arising from non-trading book activities.* EBA/GL/2022/14. https://www.eba.europa.eu/sites/default/files/document_library/Publications/Guidelines/2022/EBA-GL-2022-14%20GL%20on%20IRRBB%20and%20CSRBB/1041754/Guidelines%20on%20IRRBB%20and%20CSRBB.pdf

[9] **Entrop, O., Memmel, C., Ruprecht, B., Wilkens, M.** (2015). Determinants of bank interest margins: Impact of maturity transformation. *Journal of Banking & Finance*, 54, 1–19. https://doi.org/10.1016/j.jbankfin.2014.12.001

[10] **Fabozzi, F.J., Mann, S.V.** (2012). *The Handbook of Fixed Income Securities*, 8th edition. McGraw-Hill.

[11] **FDIC** (1997). *History of the Eighties, Lessons for the Future.* https://www.fdic.gov/bank/historical/history/

[12] **Goodhart, C.A.E.** (1975). Problems of monetary management: The UK experience. *Papers in Monetary Economics*, Reserve Bank of Australia.

[13] **Guttentag, J., Herring, R.** (1986). *Disaster Myopia in International Banking.* Essays in International Finance, Princeton University.

[14] **Hull, J., White, A.** (1990). Pricing interest-rate-derivative securities. *Review of Financial Studies*, 3(4), 573–592. https://doi.org/10.1093/rfs/3.4.573

[15] **IASB** (2014, rev. 2020). *IFRS 9, Financial Instruments.* International Accounting Standards Board. https://www.ifrs.org/issued-standards/list-of-standards/ifrs-9-financial-instruments/

[16] **Markowitz, H.** (1952). Portfolio selection. *Journal of Finance*, 7(1), 77–91. https://doi.org/10.1111/j.1540-6261.1952.tb01525.x

[17] **Memmel, C.** (2011). Banks' exposure to interest rate risk, their earnings from term transformation, and the dynamics of the term structure. *Journal of Banking & Finance*, 35(2), 282–289. https://doi.org/10.1016/j.jbankfin.2010.08.003

[18] **Observatoire Crédit Logement / CSA** (2024). *Monthly dashboard of the mortgage market*, January 2024. www.lobservatoirecreditlogement.fr

[19] **Richard, S.F., Roll, R.** (1989). Prepayments on fixed-rate mortgage-backed securities. *Journal of Portfolio Management*, 15(3), 73–82.

[20] **Sharpe, S.A.** (1990). Asymmetric information, bank lending and implicit contracts: A stylized model of customer relationships. *Journal of Finance*, 45(4), 1069–1087. https://doi.org/10.1111/j.1540-6261.1990.tb02427.x

[21] **Société Générale** (2024). *Document d'enregistrement universel 2024*, section 4.6 Global interest rate risk, p. 312. https://www.societegenerale.com/sites/default/files/documents/2024-03/document-enregistrement-universel-2024.pdf